॥वसुधैव कुटुम्बकम्॥

Symbiosis Institute of Technology, Pune

Faculty of Engineering

CSE - Academic Year 2026-27

Java Programming – Lab (Batch 2025-29)

Lab Assignment No:- 23

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Batch	2025-29
Class	Programming with JAVA
Academic Year & Semester	2nd Year – Sem III
Title of Assignment	Develop a Java application using JDBC to navigate and retrieve records from a ResultSet.
Practice questions	1. Create a program to display database records one by one using ResultSet.

```
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.ResultSet;
import java.sql.SQLException;
import java.sql.Statement;

public class RecordByRecordDemo {
    public static void main(String[] args) {
        String url = "jdbc:mysql://localhost:3306/testdb";
        String user = "root";
        String password = "XXXXX";

        try (Connection conn = DriverManager.getConnection(url,
user, password);
             Statement stmt = conn.createStatement();
             ResultSet rs = stmt.executeQuery("SELECT roll_no,
name, course FROM students")) {

            int count = 1;
            while (rs.next()) {
```

```
                System.out.println("Record " + count + ":");
                System.out.println("  Roll No : " +
rs.getInt("roll_no"));
                System.out.println("  Name     : " +
rs.getString("name"));
                System.out.println("  Course  : " +
rs.getString("course"));
                count++;
            }

        } catch (SQLException e) {
            System.out.println("Database error: " +
e.getMessage());
        }
    }
}
```

2. Create a Java program that retrieves employee records from a database and displays them one by one using ResultSet. Show employee ID, name, department and salary sequentially.

```
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.ResultSet;
import java.sql.SQLException;
import java.sql.Statement;

public class EmployeeResultSetDemo {
    public static void main(String[] args) {
        String url = "jdbc:mysql://localhost:3306/testdb";
        String user = "root";
        String password = "XXXXX";

        try (Connection conn = DriverManager.getConnection(url,
user, password);
             Statement stmt = conn.createStatement();
             ResultSet rs = stmt.executeQuery("SELECT emp_id,
name, department, salary FROM employee_details")) {

            System.out.println("---- Employee Records----- ");
            while (rs.next()) {
                System.out.println("Employee ID : " +
rs.getInt("emp_id"));
                System.out.println("Name        : " +
rs.getString("name"));
                System.out.println("Department  : " +
rs.getString("department"));
```

```
                System.out.println("Salary      : " +
rs.getDouble("salary"));
                System.out.println();
            }

        } catch (SQLException e) {
            System.out.println("Database error: " +
e.getMessage());
        }
    }
}
```

Output

1)

```
C:\Users\bkor\OneDrive\Desktop\JAVA_PRACTICALS\JAVA_EXP_23>j
Record 1:
  Roll No : 3
  Name    : Vikram Singh
  Course  : Computer Engineering
Record 2:
  Roll No : 4
  Name    : Priya Nair
  Course  : Mechanical Engineering
```

2)

```
C:\Users\bkor\OneDrive\Desktop\JAVA_PRACTICALS\JAVA_EXP_23>java -cp .;mysql-connector
---  Employee Records ----
Employee ID : 1
Name        : Rohit Sharma
Department  : Sales
Salary      : 55000.0

Employee ID : 2
Name        : Sneha Patil
Department  : HR
Salary      : 48000.0

Employee ID : 3
Name        : Kunal Mehta
Department  : IT
Salary      : 62000.0
```

Conclusion

A ResultSet works like a cursor over the rows a query returns. The default one only moves forward with next(), while a scrollable one can jump to any row. Together with the getXxx() methods for reading column values, it is how Java programs read and display database records.